

OrderCore - Technical Specification

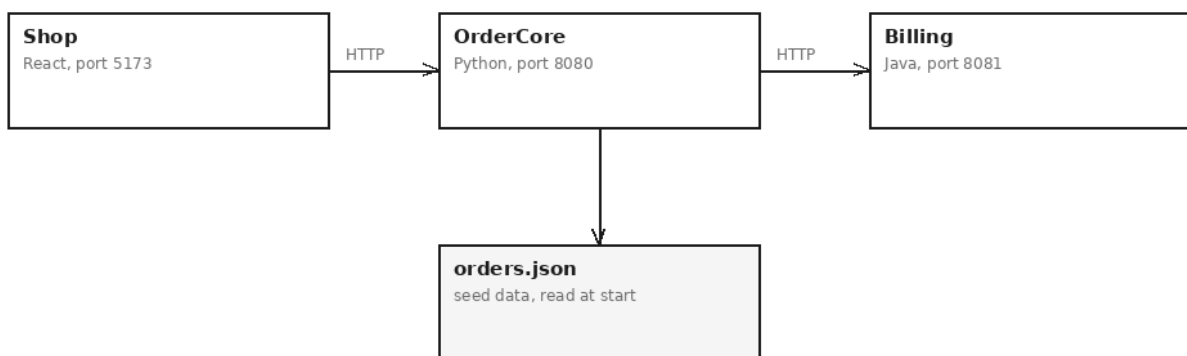
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Companion to the functional specification. This document describes how OrderCore is built and run. Where the two disagree, the functional specification describes intent and this document describes what is deployed; raise the difference rather than resolving it silently.

1. Runtime shape

Three processes, started by hand in the training environment and by the pipeline elsewhere. There is no orchestration layer, no service discovery and no shared database. Each process holds its own state in memory and reads its seed data from disk at start.

Figure 1 - Deployment, single environment



Billing call timeout: 2000 ms. On timeout OrderCore falls back to a local calculation and marks the invoice source as 'fallback'.

No database. No message queue. Restarting OrderCore loses runtime orders.

Figure 1. Ports, the Billing timeout and the fallback rule are recorded here and nowhere else.

2. The invoice path

The only cross-language call in the estate. OrderCore translates its snake_case fields to Billing's camelCase on the way out and back again on the way in. That translation is the contract, and it is enforced by nothing but code review.

Figure 2 - Invoice request, success and fallback

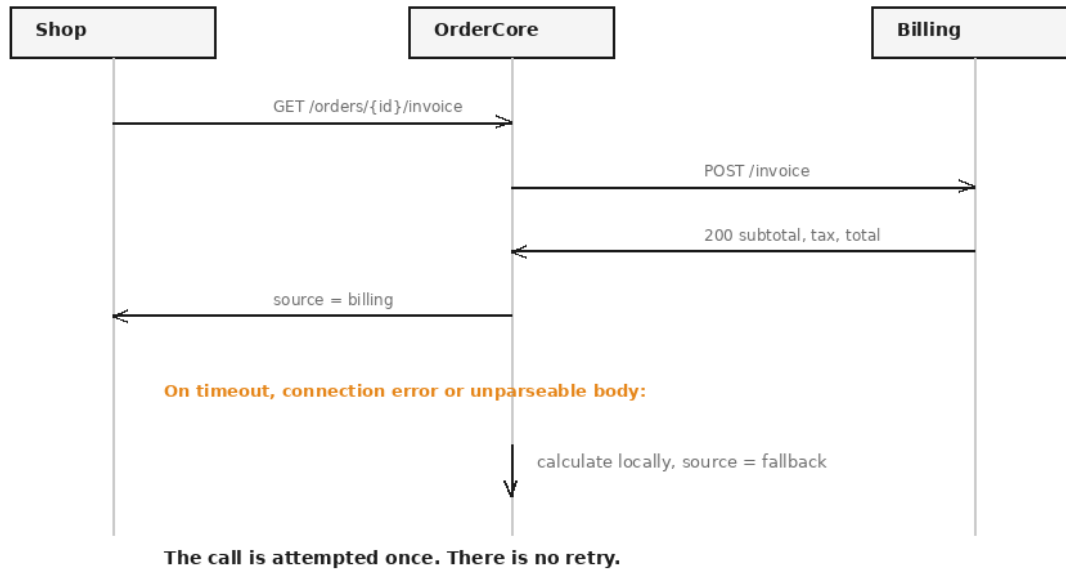


Figure 2. The retry behaviour is recorded in this diagram and nowhere else in the document.

2.1 Field mapping

OrderCore	Billing	Type	Notes
order_id	orderId	string	Echoed back unchanged
unit_price_pence	unitPricePence	integer	Whole pence, never a decimal
discount_percent	discountPercent	integer	0 to 50 inclusive, added HEL-207
subtotal_pence	subtotalPence	integer	Derived, not stored
tax_pence	taxPence	integer	20 percent, calculated once on the order
total_pence	totalPence	integer	subtotal minus discount, plus tax

3. Error handling

Three responses leave the service, and one condition that looks like an error to an operator is not an error to the caller.

Figure 3 - Error responses

Condition	Status	Body	Logged
Order id not found	404	detail: Order <id> not found	info
Validation failure	422	field, rule, value received	info
Billing unreachable	200	invoice, source = fallback	warning
Billing returns 5xx	200	invoice, source = fallback	warning
Unexpected exception	500	empty	error with stack trace

A fallback invoice is never an error to the caller. This is deliberate and is the single most common source of misplaced confidence in a green test run.

Figure 3. The error table exists only as an image in this document.

4. Performance

Measured in staging against the seeded dataset. The figures below are the only measurements anyone has taken since the service went live.

Figure 4 - Observed response times, staging, 2026-02

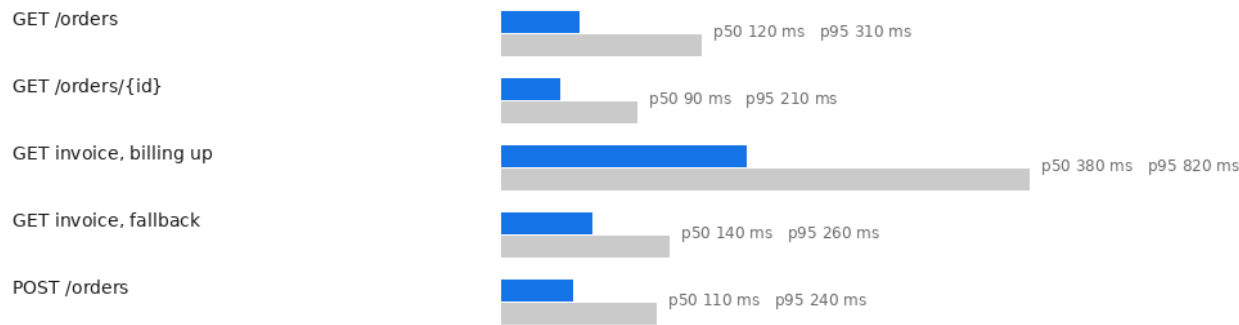


Figure 4. One of these bars breaches a stated non-functional requirement.

5. Configuration

Setting	Where it lives	Default	Changing it
Billing base URL	billing_client.py	http://localhost:8081	Code change and review
Billing timeout	billing_client.py	2000 ms	Code change and review
Tax rate	Both sides	20 percent	Architect approval, both copies
Max discount	Both sides	50 percent	Architect approval, both copies
Seed file	orders.json	five orders	Never edited to pass a test

Two of these values are written down in two languages. There is no mechanism that checks the copies agree; the only protection is that both changes appear in the same review.

6. Testing

Suite	Command	Covers	Does not cover
OrderCore	python -m pytest apps/ordercore/tests	Validation, lifecycle, the fallback path	Anything that reaches Billing over the wire
Billing	JUnit, no runner installed	Calculation, request parsing	What OrderCore actually sends
Lint	python -m ruff check apps/ordercore	Style, unused imports	Correctness

7. Known gaps

Ref	Gap	Consequence	Owner
TG-1	No contract test across the wire	A field can be dropped in transit unnoticed	Architect
TG-2	Tax rate duplicated in two languages	Silent divergence	Architect
TG-3	Invoice p95 over the NFR since February	Undetected breach	Developer
TG-4	Runtime orders lost on restart	Acceptable in training only	Product
TG-5	No caller identity on any request	NFR-7 cannot be met	Architect

This table is the most useful page in the document and the least read, because it is the last one and nothing links to it.